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第三章 不定积分

§ 1. 最简单的不定积分

1° 不定积分的概念 若函数 $f(x)$ 在区间 (a, b) 内有定义且连续, $F(x)$ 是它的原函数, 即当 $a < x < b$ 时 $F'(x) = f(x)$, 则

$$\int f(x) dx = F(x) + C, \quad a < x < b,$$

式中 C 为任意常数.

2° 不定积分的基本性质:

$$(1) d\left[\int f(x) dx\right] = f(x) dx;$$

$$(2) \int d\Phi(x) = \Phi(x) + C;$$

$$(3) \int A f(x) dx = A \int f(x) dx \quad (A \text{ 为常数}, A \neq 0);$$

$$(4) \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx.$$

3° 最简积分表:

$$\text{I. } \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1);$$

$$\text{II. } \int \frac{dx}{x} = \ln|x| + C \quad (x \neq 0);$$

$$\text{III. } \int \frac{dx}{1+x^2} = \begin{cases} \arctan x + C, \\ -\operatorname{arccot} x + C; \end{cases}$$

$$\text{IV. } \int \frac{dx}{1-x^2} = \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C;$$

$$\text{V. } \int \frac{dx}{\sqrt{1-x^2}} = \begin{cases} \arcsin x + C, \\ -\arccos x + C; \end{cases}$$

$$\text{VI. } \int \frac{dx}{\sqrt{x^2 \pm 1}} = \ln|x + \sqrt{x^2 \pm 1}| + C;$$

$$\text{VII. } \int a^x dx = \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1); \quad \int e^x dx = e^x + C;$$

$$\text{VIII. } \int \sin x dx = -\cos x + C;$$

$$\text{IX. } \int \cos x dx = \sin x + C;$$

$$\text{X. } \int \frac{dx}{\sin^2 x} = -\cot x + C;$$

$$\text{XI. } \int \frac{dx}{\cos^2 x} = \tan x + C;$$

$$\text{XII. } \int \operatorname{sh} x dx = \operatorname{ch} x + C;$$

$$\text{XIII. } \int \operatorname{ch} x dx = \operatorname{sh} x + C;$$

$$\text{XIV. } \int \frac{dx}{\operatorname{sh}^2 x} = -\operatorname{coth} x + C;$$

$$\text{XV. } \int \frac{dx}{\operatorname{ch}^2 x} = \operatorname{th} x + C.$$

4° 积分的基本方法

$$(1) \text{ 引入新变量法 若 } \int f(x) dx = F(x) + C,$$

则 $\int f(u) du = F(u) + C$, 式中 $u = \varphi(x)$ 是连续可微函数.

$$(2) \text{ 分项积分法 若 } f(x) = f_1(x) + f_2(x),$$

$$\text{则 } \int f(x) dx = \int f_1(x) dx + \int f_2(x) dx.$$

$$(3) \text{ 代入法 若 } f(x) \text{ 连续, 令 } x = \varphi(t), \text{ 式中 } \varphi(t) \text{ 及其导数 } \varphi'(t) \text{ 皆连续,}$$

$$\text{则得 } \int f(x) dx = \int f[\varphi(t)] \varphi'(t) dt.$$

$$(4) \text{ 分部积分法 若 } u \text{ 和 } v \text{ 为 } x \text{ 的可微函数, 则 } \int u dv = uv - \int v du.$$

利用最简积分表,求下列积分*:

【1628】 $\int (3-x^2)^3 dx.$

解 $\int (3-x^2)^3 dx = \int (27-27x^2+9x^4-x^6) dx = 27x-9x^3+\frac{9}{5}x^5-\frac{1}{7}x^7+C.$

【1629】 $\int x^2(5-x)^4 dx.$

解 $\int x^2(5-x)^4 dx = \int (625x^2-500x^3+150x^4-20x^5+x^6) dx = \frac{625}{3}x^3-125x^4+30x^5-\frac{10}{3}x^6+\frac{1}{7}x^7+C.$

【1630】 $\int (1-x)(1-2x)(1-3x) dx.$

解 $\int (1-x)(1-2x)(1-3x) dx = \int (1-6x+11x^2-6x^3) dx = x-3x^2+\frac{11}{3}x^3-\frac{3}{2}x^4+C.$

【1631】 $\int \left(\frac{1-x}{x}\right)^2 dx.$

解 $\int \left(\frac{1-x}{x}\right)^2 dx = \int \left(\frac{1}{x^2}-\frac{2}{x}+1\right) dx = -\frac{1}{x}-2\ln|x|+x+C.$

【1632】 $\int \left(\frac{a}{x}+\frac{a^2}{x^2}+\frac{a^3}{x^3}\right) dx.$

解 $\int \left(\frac{a}{x}+\frac{a^2}{x^2}+\frac{a^3}{x^3}\right) dx = a\ln|x|-\frac{a^2}{x}-\frac{a^3}{2x^2}+C.$

【1633】 $\int \frac{x+1}{\sqrt{x}} dx.$

解 $\int \frac{x+1}{\sqrt{x}} dx = \int (x^{\frac{1}{2}}+x^{-\frac{1}{2}}) dx = \frac{2}{3}x\sqrt{x}+2\sqrt{x}+C.$

【1634】 $\int \frac{\sqrt{x}-2\sqrt[3]{x^2}+1}{\sqrt[4]{x}} dx.$

解 $\int \frac{\sqrt{x}-2\sqrt[3]{x^2}+1}{\sqrt[4]{x}} dx = \int (x^{\frac{1}{4}}-2x^{\frac{5}{12}}+x^{-\frac{1}{4}}) dx = \frac{4}{5}x\sqrt{x}-\frac{24}{17}x^{\frac{12}{17}}\sqrt[12]{x^5}+\frac{4}{3}\sqrt[4]{x^3}+C.$

【1635】 $\int \frac{(1-x)^3}{x\sqrt{x}} dx.$

解 $\int \frac{(1-x)^3}{x\sqrt{x}} dx = \int (x^{-\frac{4}{3}}-3x^{-\frac{1}{3}}+3x^{\frac{2}{3}}-x^{\frac{5}{3}}) dx = -\frac{3}{\sqrt[3]{x}}\left(1+\frac{3}{2}x-\frac{3}{5}x^2+\frac{1}{8}x^3\right)+C.$

【1636】 $\int \left(1-\frac{1}{x^2}\right)\sqrt{x\sqrt{x}} dx.$

提示 注意 $\left(1-\frac{1}{x^2}\right)\sqrt{x\sqrt{x}} = x^{\frac{3}{4}}-x^{-\frac{5}{4}}.$

1634 题,1635 题,1637 题及 1638 题均可仿本题,将被积函数化成若干个幂函数的代数和,然后再利用分项积分法.

解 $\int \left(1-\frac{1}{x^2}\right)\sqrt{x\sqrt{x}} dx = \int (x^{\frac{3}{4}}-x^{-\frac{5}{4}}) dx = \frac{4}{7}x^{\frac{7}{4}}+4x^{-\frac{1}{4}}+C = \frac{4(x^2+7)}{7\sqrt[4]{x}}+C.$

【1637】 $\int \frac{(\sqrt{2x}-\sqrt[3]{3x})^2}{x} dx.$

* 本章在叙述习题及其解答过程中,凡出现的函数,无论是被积函数还是原函数,均默认是在有意义的定义域上进行的.例如,最简积分表 I 中当 $n \leq -2$ 时,要求 $x \neq 0$; IV 中要求 $|x| \neq 1$; V 中要求 $|x| < 1$; 以及 VI 中,当取负号时要求 $|x| > 1$; 等等,就未加声明.在题解中也有相当多的类似情况.因此,如无特别声明,在一般情况下,这些定义域是很容易被读者确定的,此处就不再予以一一指明.

$$\text{解} \quad \int \frac{(\sqrt{2x} - \sqrt[3]{3x})^2}{x} dx = \int (2 - 2\sqrt[6]{72}x^{-\frac{1}{6}} + \sqrt[3]{9}x^{-\frac{1}{3}}) dx = 2x - \frac{12}{5}\sqrt[6]{72x^5} + \frac{3}{2}\sqrt[3]{9x^2} + C.$$

$$\text{【1638】} \quad \int \frac{\sqrt{x^4 + x^{-4} + 2}}{x^3} dx.$$

$$\text{解} \quad \int \frac{\sqrt{x^4 + x^{-4} + 2}}{x^3} dx = \int \frac{x^2 + \frac{1}{x^2}}{x^3} dx = \int \left(\frac{1}{x} + \frac{1}{x^5} \right) dx = \ln|x| - \frac{1}{4x^4} + C.$$

$$\text{【1639】} \quad \int \frac{x^2}{1+x^2} dx.$$

$$\text{解} \quad \int \frac{x^2}{1+x^2} dx = \int \left(1 - \frac{1}{x^2+1} \right) dx = x - \arctan x + C.$$

$$\text{【1640】} \quad \int \frac{x^2}{1-x^2} dx.$$

$$\text{解} \quad \int \frac{x^2}{1-x^2} dx = \int \left(-1 + \frac{1}{1-x^2} \right) dx = -x + \frac{1}{2} \ln \left| \frac{1+x}{1-x} \right| + C.$$

$$\text{【1641】} \quad \int \frac{x^2+3}{x^2-1} dx.$$

$$\text{解} \quad \int \frac{x^2+3}{x^2-1} dx = \int \left(1 + \frac{4}{x^2-1} \right) dx = x + 2 \ln \left| \frac{x-1}{x+1} \right| + C.$$

$$\text{【1642】} \quad \int \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1-x^4}} dx.$$

$$\text{解} \quad \int \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1-x^4}} dx = \int \left(\frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1+x^2}} \right) dx = \arcsin x + \ln(x + \sqrt{1+x^2}) + C.$$

$$\text{【1643】} \quad \int \frac{\sqrt{x^2+1} - \sqrt{x^2-1}}{\sqrt{x^4-1}} dx.$$

$$\text{解} \quad \int \frac{\sqrt{x^2+1} - \sqrt{x^2-1}}{\sqrt{x^4-1}} dx = \int \left(\frac{1}{\sqrt{x^2-1}} - \frac{1}{\sqrt{x^2+1}} \right) dx = \ln \left| \frac{x + \sqrt{x^2-1}}{x + \sqrt{x^2+1}} \right| + C.$$

$$\text{【1644】} \quad \int (2^x + 3^x)^2 dx.$$

$$\text{解} \quad \int (2^x + 3^x)^2 dx = \int (4^x + 2 \cdot 6^x + 9^x) dx = \frac{4^x}{\ln 4} + 2 \cdot \frac{6^x}{\ln 6} + \frac{9^x}{\ln 9} + C.$$

$$\text{【1645】} \quad \int \frac{2^{x+1} - 5^{x-1}}{10^x} dx.$$

$$\text{解} \quad \int \frac{2^{x+1} - 5^{x-1}}{10^x} dx = \int \left[2 \left(\frac{1}{5} \right)^x - \frac{1}{5} \left(\frac{1}{2} \right)^x \right] dx = -\frac{2}{\ln 5} \left(\frac{1}{5} \right)^x + \frac{1}{5 \ln 2} \left(\frac{1}{2} \right)^x + C.$$

$$\text{【1646】} \quad \int \frac{e^{3x} + 1}{e^x + 1} dx.$$

$$\text{解} \quad \int \frac{e^{3x} + 1}{e^x + 1} dx = \int (e^{2x} - e^x + 1) dx = \frac{1}{2} e^{2x} - e^x + x + C.$$

$$\text{【1647】} \quad \int (1 + \sin x + \cos x) dx.$$

$$\text{解} \quad \int (1 + \sin x + \cos x) dx = x - \cos x + \sin x + C.$$

$$\text{【1648】} \quad \int \sqrt{1 - \sin 2x} dx.$$

提示 注意 $\sqrt{1 - \sin 2x} = \sqrt{(\cos x - \sin x)^2} = [\operatorname{sgn}(\cos x - \sin x)](\cos x - \sin x)$.

$$\begin{aligned} \text{解} \quad \int \sqrt{1 - \sin 2x} dx &= \int \sqrt{(\cos x - \sin x)^2} dx = \int [\operatorname{sgn}(\cos x - \sin x)](\cos x - \sin x) dx \\ &= (\sin x + \cos x) \operatorname{sgn}(\cos x - \sin x) + C. \end{aligned}$$

【1649】 $\int \cot^2 x dx.$

提示 注意 $\cot^2 x = \csc^2 x - 1$.

解 $\int \cot^2 x dx = \int (\csc^2 x - 1) dx = -\cot x - x + C.$

【1650】 $\int \tan^2 x dx.$

解 $\int \tan^2 x dx = \int (\sec^2 x - 1) dx = \tan x - x + C.$

【1651】 $\int (a \operatorname{sh} x + b \operatorname{ch} x) dx.$

解 $\int (a \operatorname{sh} x + b \operatorname{ch} x) dx = a \operatorname{ch} x + b \operatorname{sh} x + C.$

【1652】 $\int \operatorname{th}^2 x dx.$

提示 注意 $\operatorname{th}^2 x = 1 - \frac{1}{\operatorname{ch}^2 x}$.

解 $\int \operatorname{th}^2 x dx = \int \left(1 - \frac{1}{\operatorname{ch}^2 x}\right) dx = x - \operatorname{th} x + C.$

【1653】 $\int \operatorname{coth}^2 x dx.$

提示 注意 $\operatorname{coth}^2 x = 1 + \frac{1}{\operatorname{sh}^2 x}$.

解 $\int \operatorname{coth}^2 x dx = \int \left(1 + \frac{1}{\operatorname{sh}^2 x}\right) dx = x - \operatorname{coth} x + C.$

【1654】 证明:若 $\int f(x) dx = F(x) + C$, 则 $\int f(ax+b) dx = \frac{1}{a} F(ax+b) + C$ ($a \neq 0$).

提示 由不定积分的定义, 命题即获证.

证 由 $\int f(x) dx = F(x) + C$ 得知 $F'(x) = f(x)$. 因而有 $F'(ax+b) = f(ax+b)$, 且

$$\frac{d}{dx} \left[\frac{1}{a} F(ax+b) \right] = F'(ax+b),$$

于是,

$$\frac{d}{dx} \left[\frac{1}{a} F(ax+b) \right] = f(ax+b),$$

所以,

$$\int f(ax+b) dx = \frac{1}{a} F(ax+b) + C.$$

求下列积分:

【1655】 $\int \frac{dx}{x+a}.$

解 $\int \frac{dx}{x+a} = \ln |x+a| + C.$

【1656】 $\int (2x-3)^{10} dx.$

解 $\int (2x-3)^{10} dx = \frac{1}{2} \cdot \frac{1}{11} (2x-3)^{11} + C = \frac{1}{22} (2x-3)^{11} + C.$

【1657】 $\int \sqrt[3]{1-3x} dx.$

解 $\int \sqrt[3]{1-3x} dx = -\frac{1}{3} \cdot \frac{3}{4} (1-3x)^{\frac{4}{3}} + C = -\frac{1}{4} (1-3x)^{\frac{4}{3}} + C.$

【1658】 $\int \frac{dx}{\sqrt{2-5x}}.$

$$\text{解} \quad \int \frac{dx}{\sqrt{2-5x}} = -\frac{1}{5} \cdot 2(2-5x)^{\frac{1}{2}} + C = -\frac{2}{5} \sqrt{2-5x} + C.$$

$$\text{【1659】} \quad \int \frac{dx}{(5x-2)^{\frac{5}{2}}}.$$

$$\text{解} \quad \int \frac{dx}{(5x-2)^{\frac{5}{2}}} = \frac{1}{5} \cdot \left(-\frac{2}{3}\right) (5x-2)^{-\frac{3}{2}} + C = -\frac{2}{15(5x-2)^{\frac{3}{2}}} + C.$$

$$\text{【1660】}^{+} \quad \int \frac{\sqrt{1-2x+x^2}}{1-x} dx.$$

$$\text{解} \quad \int \frac{\sqrt{1-2x+x^2}}{1-x} dx = \int (1-x)^{-\frac{3}{2}} dx = -\frac{5}{2} \sqrt{(1-x)^2} + C.$$

$$\text{【1661】} \quad \int \frac{dx}{2+3x^2}.$$

$$\text{解} \quad \int \frac{dx}{2+3x^2} = \int \frac{dx}{(\sqrt{2})^2 + (\sqrt{3}x)^2} = \frac{1}{\sqrt{6}} \arctan\left(x\sqrt{\frac{3}{2}}\right) + C.$$

$$\text{【1662】} \quad \int \frac{dx}{2-3x^2}.$$

$$\text{解} \quad \int \frac{dx}{2-3x^2} = \frac{1}{2} \int \frac{dx}{1 - \left(\sqrt{\frac{3}{2}}x\right)^2} = \frac{1}{2} \cdot \sqrt{\frac{2}{3}} \cdot \frac{1}{2} \ln \left| \frac{1 + \sqrt{\frac{3}{2}}x}{1 - \sqrt{\frac{3}{2}}x} \right| + C = \frac{1}{2\sqrt{6}} \ln \left| \frac{\sqrt{2} + x\sqrt{3}}{\sqrt{2} - x\sqrt{3}} \right| + C.$$

$$\text{【1663】} \quad \int \frac{dx}{\sqrt{2-3x^2}}.$$

$$\text{解} \quad \int \frac{dx}{\sqrt{2-3x^2}} = \frac{1}{\sqrt{3}} \arcsin\left(x\sqrt{\frac{3}{2}}\right) + C.$$

$$\text{【1664】} \quad \int \frac{dx}{\sqrt{3x^2-2}}.$$

$$\begin{aligned} \text{解} \quad \int \frac{dx}{\sqrt{3x^2-2}} &= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\left(\sqrt{\frac{3}{2}}x\right)^2 - 1}} = \frac{1}{\sqrt{2}} \cdot \sqrt{\frac{2}{3}} \ln \left| x\sqrt{\frac{3}{2}} + \sqrt{\frac{3}{2}x^2 - 1} \right| + C_1 \\ &= \frac{1}{\sqrt{3}} \ln |x\sqrt{3} + \sqrt{3x^2 - 2}| + C. \end{aligned}$$

$$\text{【1665】} \quad \int (e^{-x} + e^{-2x}) dx.$$

$$\text{解} \quad \int (e^{-x} + e^{-2x}) dx = -\left(e^{-x} + \frac{1}{2}e^{-2x}\right) + C.$$

$$\text{【1666】} \quad \int (\sin 5x - \sin 5a) dx.$$

$$\text{解} \quad \int (\sin 5x - \sin 5a) dx = -\frac{1}{5} \cos 5x - x \sin 5a + C.$$

$$\text{【1667】} \quad \int \frac{dx}{\sin^2\left(2x + \frac{\pi}{4}\right)}.$$

$$\text{解} \quad \int \frac{dx}{\sin^2\left(2x + \frac{\pi}{4}\right)} = -\frac{1}{2} \cot\left(2x + \frac{\pi}{4}\right) + C$$

* 题号右上角带“+”号表示题解答案与原习题集中译本所附答案不一致,以后不再说明.中译本基本是按俄文第二版翻译的.俄文第二版中有一些错误已在俄文第三版中改正.

【1668】 $\int \frac{dx}{1+\cos x}.$

解 $\int \frac{dx}{1+\cos x} = \frac{1}{2} \int \frac{dx}{\cos^2 \frac{x}{2}} = \tan \frac{x}{2} + C.$

【1669】 $\int \frac{dx}{1-\cos x}.$

解 $\int \frac{dx}{1-\cos x} = \frac{1}{2} \int \frac{dx}{\sin^2 \frac{x}{2}} = -\cot \frac{x}{2} + C.$

【1670】 $\int \frac{dx}{1+\sin x}.$

提示 注意 $\frac{1}{1+\sin x} = \frac{1}{1+\cos(\frac{\pi}{2}-x)}$, 并利用 1668 题的结果.

解 $\int \frac{dx}{1+\sin x} = \int \frac{dx}{1+\cos(\frac{\pi}{2}-x)} = -\tan(\frac{\pi}{4}-\frac{x}{2}) + C.$

【1671】 $\int [\operatorname{sh}(2x+1) + \operatorname{ch}(2x-1)] dx.$

解 $\int [\operatorname{sh}(2x+1) + \operatorname{ch}(2x-1)] dx = \frac{1}{2} [\operatorname{ch}(2x+1) + \operatorname{sh}(2x-1)] + C.$

【1672】 $\int \frac{dx}{\operatorname{ch}^2 \frac{x}{2}}.$

解 $\int \frac{dx}{\operatorname{ch}^2 \frac{x}{2}} = 2 \operatorname{th} \frac{x}{2} + C.$

【1673】 $\int \frac{dx}{\operatorname{sh}^2 \frac{x}{2}}.$

解 $\int \frac{dx}{\operatorname{sh}^2 \frac{x}{2}} = -2 \operatorname{coth} \frac{x}{2} + C.$

用适当地变换被积函数的方法求下列积分:

【1674】 $\int \frac{x dx}{\sqrt{1-x^2}}.$

解 $\int \frac{x dx}{\sqrt{1-x^2}} = -\int \frac{d(1-x^2)}{2\sqrt{1-x^2}} = -\sqrt{1-x^2} + C.$

【1675】 $\int x^2 \sqrt[3]{1+x^3} dx.$

提示 注意 $x^2 \sqrt[3]{1+x^3} dx = \frac{1}{3} (1+x^3)^{\frac{1}{3}} d(1+x^3).$

解 $\int x^2 \sqrt[3]{1+x^3} dx = \frac{1}{3} \int (1+x^3)^{\frac{1}{3}} d(1+x^3) = \frac{1}{4} (1+x^3)^{\frac{4}{3}} + C.$

【1676】 $\int \frac{x dx}{3-2x^2}.$

解 $\int \frac{x dx}{3-2x^2} = -\frac{1}{4} \int \frac{d(3-2x^2)}{3-2x^2} = -\frac{1}{4} \ln |3-2x^2| + C.$

【1677】 $\int \frac{x dx}{(1+x^2)^2}.$

解 $\int \frac{x dx}{(1+x^2)^2} = \frac{1}{2} \int \frac{d(1+x^2)}{(1+x^2)^2} = -\frac{1}{2(1+x^2)} + C.$

【1678】 $\int \frac{x dx}{4+x^4}.$

提示 注意 $\frac{x dx}{4+x^4} = \frac{1}{2} \cdot \frac{d(x^2)}{2^2+(x^2)^2}.$

解 $\int \frac{x dx}{4+x^4} = \frac{1}{2} \int \frac{d(x^2)}{2^2+(x^2)^2} = \frac{1}{4} \arctan \frac{x^2}{2} + C.$

【1679】 $\int \frac{x^3 dx}{x^8-2}.$

解 $\int \frac{x^3 dx}{x^8-2} = \frac{1}{4} \int \frac{d(x^4)}{(x^4)^2-(\sqrt{2})^2} = \frac{1}{8\sqrt{2}} \ln \left| \frac{x^4-\sqrt{2}}{x^4+\sqrt{2}} \right| + C.$

【1680】 $\int \frac{dx}{\sqrt{x}(1+x)}.$

提示 注意 $\frac{dx}{\sqrt{x}(1+x)} = 2 \cdot \frac{d(\sqrt{x})}{1+(\sqrt{x})^2}.$

解 $\int \frac{dx}{\sqrt{x}(1+x)} = 2 \int \frac{d(\sqrt{x})}{1+(\sqrt{x})^2} = 2 \arctan \sqrt{x} + C.$

【1681】 $\int \sin \frac{1}{x} \cdot \frac{dx}{x^2}.$

解 $\int \sin \frac{1}{x} \cdot \frac{dx}{x^2} = - \int \sin \frac{1}{x} d\left(\frac{1}{x}\right) = \cos \frac{1}{x} + C.$

【1682】 $\int \frac{dx}{x \sqrt{x^2+1}}.$

提示 注意 $\frac{dx}{x \sqrt{x^2+1}} = \frac{dx}{x|x| \sqrt{1+\frac{1}{x^2}}} = -\frac{d\left(\frac{1}{|x|}\right)}{\sqrt{1+\left(\frac{1}{|x|}\right)^2}}.$

解 $\int \frac{dx}{x \sqrt{x^2+1}} = \int \frac{dx}{x|x| \sqrt{1+\frac{1}{x^2}}} = - \int \frac{d\left(\frac{1}{|x|}\right)}{\sqrt{1+\left(\frac{1}{|x|}\right)^2}} = -\ln\left(\frac{1}{|x|} + \sqrt{1+\frac{1}{x^2}}\right) + C$
 $= -\ln \left| \frac{1+\sqrt{x^2+1}}{x} \right| + C.$

【1683】 $\int \frac{dx}{x \sqrt{x^2-1}}.$

提示 仿 1682 题的解法.

解 $\int \frac{dx}{x \sqrt{x^2-1}} = \int \frac{dx}{x|x| \sqrt{1-\frac{1}{x^2}}} = - \int \frac{d\left(\frac{1}{|x|}\right)}{\sqrt{1-\left(\frac{1}{|x|}\right)^2}} = -\arcsin \frac{1}{|x|} + C.$

【1684】 $\int \frac{dx}{(x^2+1)^{\frac{3}{2}}}.$

提示 注意 $\frac{dx}{(x^2+1)^{\frac{3}{2}}} = \frac{\operatorname{sgn} x dx}{x^3 \left(1+\frac{1}{x^2}\right)^{\frac{3}{2}}} = -\frac{1}{2} \left(1+\frac{1}{x^2}\right)^{-\frac{3}{2}} \operatorname{sgn} x d\left(1+\frac{1}{x^2}\right).$

解 $\int \frac{dx}{(x^2+1)^{\frac{3}{2}}} = \int \frac{\operatorname{sgn} x dx}{x^3 \left(1+\frac{1}{x^2}\right)^{\frac{3}{2}}} = -\frac{1}{2} \int \left(1+\frac{1}{x^2}\right)^{-\frac{3}{2}} \operatorname{sgn} x d\left(1+\frac{1}{x^2}\right)$

$$= \left(1 + \frac{1}{x^2}\right)^{-\frac{1}{2}} \operatorname{sgn} x + C = \frac{x}{\sqrt{x^2+1}} + C.$$

【1685】 $\int \frac{x dx}{(x^2-1)^{\frac{3}{2}}}.$

解 $\int \frac{x dx}{(x^2-1)^{\frac{3}{2}}} = \frac{1}{2} \int (x^2-1)^{-\frac{3}{2}} d(x^2-1) = -\frac{1}{\sqrt{x^2-1}} + C.$

【1686】 $\int \frac{x^2 dx}{(8x^3+27)^{\frac{2}{3}}}.$

解 $\int \frac{x^2 dx}{(8x^3+27)^{\frac{2}{3}}} = \frac{1}{24} \int (8x^3+27)^{-\frac{2}{3}} d(8x^3+27) = \frac{1}{8} \sqrt[3]{8x^3+27} + C.$

【1687】 $\int \frac{dx}{\sqrt{x(1+x)}}.$

提示 分别就 $x > 0$ 及 $x < -1$ 时求解,然后将这两个结果合并,其结果为

$$\int \frac{dx}{\sqrt{x(1+x)}} = 2 \operatorname{sgn} x \ln(\sqrt{|x|} + \sqrt{|1+x|}) + C.$$

解 由 $x(1+x) > 0$ 知: $x > 0$ 或 $x < -1$. 当 $x > 0$ 时,

$$\int \frac{dx}{\sqrt{x(1+x)}} = 2 \int \frac{d(\sqrt{x})}{\sqrt{1+(\sqrt{x})^2}} = 2 \ln(\sqrt{x} + \sqrt{1+x}) + C;$$

当 $x < -1$ 时, $\int \frac{dx}{\sqrt{x(1+x)}} = - \int \frac{d(-(1+x))}{\sqrt{(-x)(-(1+x))}} = -2 \int \frac{d(\sqrt{-(1+x)})}{\sqrt{1+(\sqrt{-(1+x)})^2}}$
 $= -2 \ln(\sqrt{-x} + \sqrt{-(1+x)}) + C.$

总之,得 $\int \frac{dx}{\sqrt{x(1+x)}} = 2 \operatorname{sgn} x \ln(\sqrt{|x|} + \sqrt{|1+x|}) + C.$

【1688】 $\int \frac{dx}{\sqrt{x(1-x)}}.$

解 由 $x(1-x) > 0$ 知: $0 < x < 1$. 于是,得

$$\int \frac{dx}{\sqrt{x(1-x)}} = 2 \int \frac{d(\sqrt{x})}{\sqrt{1-(\sqrt{x})^2}} = 2 \arcsin \sqrt{x} + C.$$

【1689】 $\int x e^{-x^2} dx.$

解 $\int x e^{-x^2} dx = -\frac{1}{2} \int e^{-x^2} d(-x^2) = -\frac{1}{2} e^{-x^2} + C.$

【1690】 $\int \frac{e^x dx}{2+e^x}.$

解 $\int \frac{e^x dx}{2+e^x} = \int \frac{d(2+e^x)}{2+e^x} = \ln(2+e^x) + C.$

【1691】 $\int \frac{dx}{e^x + e^{-x}}.$

提示 注意 $\frac{dx}{e^x + e^{-x}} = \frac{d(e^x)}{1+(e^x)^2}.$

解 $\int \frac{dx}{e^x + e^{-x}} = \int \frac{d(e^x)}{1+(e^x)^2} = \arctan(e^x) + C.$

【1692】 $\int \frac{dx}{\sqrt{1+e^{2x}}}.$

提示 注意 $\frac{dx}{\sqrt{1+e^{2x}}} = -\frac{d(e^{-x})}{\sqrt{1+(e^{-x})^2}}.$

$$\text{解} \quad \int \frac{dx}{\sqrt{1+e^{2x}}} = - \int \frac{d(e^{-x})}{\sqrt{1+(e^{-x})^2}} = -\ln(e^{-x} + \sqrt{1+e^{-2x}}) + C.$$

$$\text{【1693】} \quad \int \frac{\ln^2 x}{x} dx.$$

$$\text{解} \quad \int \frac{\ln^2 x}{x} dx = \int \ln^2 x d(\ln x) = \frac{1}{3} \ln^3 x + C.$$

$$\text{【1694】} \quad \int \frac{dx}{x \ln x \ln(\ln x)}.$$

$$\text{解} \quad \int \frac{dx}{x \ln x \ln(\ln x)} = \int \frac{d(\ln x)}{\ln x \ln(\ln x)} = \int \frac{d[\ln(\ln x)]}{\ln(\ln x)} = \ln |\ln(\ln x)| + C.$$

$$\text{【1695】} \quad \int \sin^5 x \cos x dx.$$

$$\text{解} \quad \int \sin^5 x \cos x dx = \int \sin^4 x d(\sin x) = \frac{1}{6} \sin^6 x + C.$$

$$\text{【1696】} \quad \int \frac{\sin x}{\sqrt{\cos^3 x}} dx.$$

$$\text{解} \quad \int \frac{\sin x}{\sqrt{\cos^3 x}} dx = - \int (\cos x)^{-\frac{3}{2}} d(\cos x) = \frac{2}{\sqrt{\cos x}} + C.$$

$$\text{【1697】} \quad \int \tan x dx.$$

$$\text{解} \quad \int \tan x dx = \int \frac{\sin x}{\cos x} dx = - \int \frac{d(\cos x)}{\cos x} = -\ln |\cos x| + C.$$

$$\text{【1698】} \quad \int \cot x dx.$$

$$\text{解} \quad \int \cot x dx = \int \frac{\cos x}{\sin x} dx = \int \frac{d(\sin x)}{\sin x} = \ln |\sin x| + C.$$

$$\text{【1699】} \quad \int \frac{\sin x + \cos x}{\sqrt[3]{\sin x - \cos x}} dx.$$

$$\begin{aligned} \text{解} \quad \int \frac{\sin x + \cos x}{\sqrt[3]{\sin x - \cos x}} dx &= \int (\sin x - \cos x)^{-\frac{1}{3}} d(\sin x - \cos x) = \frac{3}{2} \sqrt[3]{(\sin x - \cos x)^2} + C \\ &= \frac{3}{2} \sqrt[3]{1 - \sin 2x} + C. \end{aligned}$$

$$\text{【1700】}^+ \quad \int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx.$$

提示 分别就 $|a| = |b| \neq 0$ 及 $|a| \neq |b|$ 两种情况求解.

解 当 $|a| = |b| \neq 0$ 时,

$$\int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx = \frac{1}{|a|} \int \sin x \cos x dx = \frac{1}{2|a|} \sin^2 x + C;$$

当 $|a| \neq |b|$ 时,

$$\begin{aligned} \int \frac{\sin x \cos x}{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}} dx &= \frac{1}{2} \int \frac{d(\sin^2 x)}{\sqrt{(a^2 - b^2) \sin^2 x + b^2}} = \frac{1}{a^2 - b^2} \sqrt{(a^2 - b^2) \sin^2 x + b^2} + C \\ &= \frac{\sqrt{a^2 \sin^2 x + b^2 \cos^2 x}}{a^2 - b^2} + C. \end{aligned}$$

$$\text{【1701】} \quad \int \frac{dx}{\sin^2 x \sqrt[4]{\cot x}}.$$

$$\text{解} \quad \int \frac{dx}{\sin^2 x \sqrt[4]{\cot x}} = - \int (\cot x)^{-\frac{1}{4}} d(\cot x) = -\frac{4}{3} \sqrt[4]{\cot^3 x} + C.$$

$$\text{【1702】} \quad \int \frac{dx}{\sin^2 x + 2 \cos^2 x}.$$